Lab 10 CheckIn – Oliver Wuttke (WUTT0019)

Question 1 – Progress

What did you accomplish this week? Any issues slowing down progress?

Final lab done, nothing much slowing down on that front. Main things for this week was working on A2, after having a group meeting we've narrowed down our machine learning architecture but still need to define our knowledge base and what our final output maybe. For my part of this I've looked into how NSW public holidays have affected demand and on average the demand is 10-15% lower on public holidays than the average for that day of the week. This is backed up by one of the papers I cited in A1 which used an ANN with fuzzy logic for short-term load forecasting for Korean power stations with some of the those logic rules being for public holidays lower demand and therefore less load, something which the network failed to capture but hardcoded rules will never miss.

Question 2 – Plans

What are your goals for next week? How does this match against your overall project plan?

Doing my part of the code and finishing up A2 (report, code, and video). I think the coding won't be too much given ML libraries abstract so much. Lines up well with the goal of finishing the project on time.

Question 3 – Study

List and discuss the three main points of material covered this week.

1. Convolutional Neural Networks – We investigated how convolutional NN's are use images are features and perform, convolutions, pooling, and padding to alter the dimensions of the images so a fully connected layer can be computationally feasible.
2. Prolog Expert Systems – In the lab we looked at using if-then-else and other logical operators in Prolog to build small scope expert systems.
3. AI Safety – Looked into recent incidents involving AI breaking laws and breaking into places and discussed how frontier labs should move forward with testing and safety.

Question 4 – Insights

What was the single most confusing concept (“muddiest point”) for you in this week’s material, and what specific question do you have about it?

The CNN’s were a bit confusing again and the idea of fully connected layers only properly clicked for me during the lecture (after already done a NN topic last year…).

Question 5 – Linkage

How could one key idea from this week’s material be directly applied to your project?

Again, it’s the Prolog since as a team we are looking into what facts and rules need to be in our expert system to provide the best output, so the more Prolog experience the better.